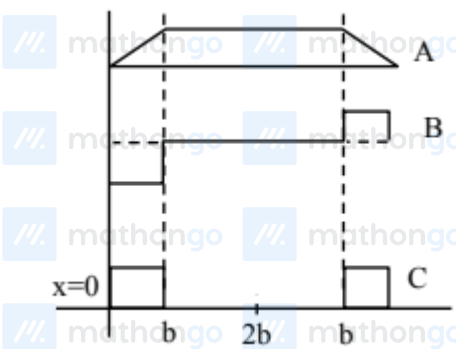
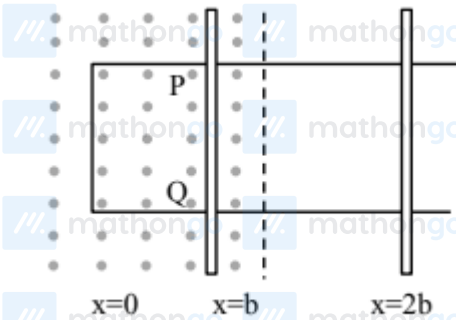


Q1 (20 July 2021 Shift 1)

The arm PQ of a rectangular conductor is moving from $x = 0$ to $x = 2b$ outwards and then inwards from $x = 2b$ to $x = 0$ as shown in the figure. A uniform magnetic field perpendicular to the plane is acting from $x = 0$ to $x = b$. Identify the graph showing the variation of different quantities with distance :



- (1) A-Flux, B-Power dissipated, C-EMF
- (2) A-Power dissipated, B-Flux, C-EMF
- (3) A-Flux, B-EMF, C-Power dissipated
- (4) A-EMF, B-Power dissipated, C-Flux

Q2 (25 July 2021 Shift 1)

A circular conducting coil of radius 1 m is being heated by the change of magnetic field $\vec{B}$ passing perpendicular to the plane in which the coil is laid. The resistance of the coil is $2\mu\Omega$. The magnetic field is slowly switched off such that its magnitude changes in time as $B = \frac{4}{\pi} \times 10^{-3} \text{ T} \left(1 - \frac{t}{100}\right)$. The energy dissipated by the coil before the magnetic field is switched off completely is $E = \text{mJ}$.

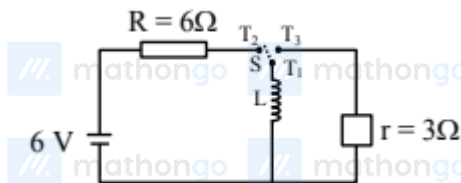
Q3 (27 July 2021 Shift 1)

Questions with Answer Keys

MathonGo

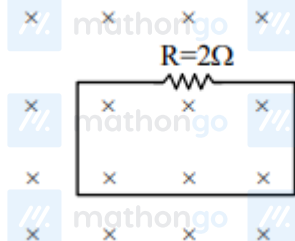
Consider an electrical circuit containing a two way switch 'S'. Initially S is open and then T_1 is connected to T_2 . As the current in $R = 6\Omega$ attains a maximum value of steady state level, T_1 is disconnected from T_2 and immediately connected to T_3 . Potential drop across $r = 3\Omega$ resistor immediately after T_1 is connected to T_3 is _____ V.

(Round off to the Nearest Integer)



Q4 (27 July 2021 Shift 2)

In the given figure the magnetic flux through the loop increases according to the relation $\phi_B(t) = 10t^2 + 20t$, where ϕ_B is in milliwebers and t is in seconds. The magnitude of current through $R = 2\Omega$ resistor at $t = 5$ s is _____ mA



Q4 (60)

Hints and Solutions

MathonGo

Q1

As rod moves in field area increases upto $\bar{x} = b$ then field is absent and again flux is generated on return journey from $x = b$ to $x = 0$. Thus plot A for flux.

$$\Rightarrow e = -\frac{d\phi}{dt} \Rightarrow \text{curve B for emf}$$

$$\Rightarrow \text{Power dissipated} = vi \Rightarrow \text{curve C for power dissipated}$$

Q2

$$\phi = \vec{B} \cdot \vec{S}$$

$$\phi = \frac{4}{\pi} \times 10^{-3} \left(1 - \frac{t}{100}\right) \cdot \pi R^2$$

$$\phi = 4 \times 10^{-3} \times (1)^2 \left(1 - \frac{t}{100}\right)$$

$$\varepsilon = -\frac{d\phi}{dt}$$

$$\varepsilon = -\frac{d}{dt} \left(4 \times 10^{-3} \left(1 - \frac{t}{100}\right)\right)$$

$$\varepsilon = 4 \times 10^{-3} \left(\frac{1}{100}\right) = 4 \times 10^{-5} \text{ V}$$

$$\text{When } B = 0 \quad 1 - \frac{t}{100} = 0$$

$$t = 100 \text{ sec}$$

$$\text{Heat} = \frac{\varepsilon^2}{R} t$$

$$\text{Heat} = \frac{(4 \times 10^{-5})^2}{2 \times 10^{-6}} \times 100 \text{ J}$$

$$\text{Heat} = \frac{16 \times 10^{-10} \times 100}{2 \times 10^{-6}} \text{ J}$$

$$\text{Heat} = 0.08 \text{ J}$$

$$\text{Heat} = 80 \text{ mJ}$$

Q3

When T_1 and T_2 are connected, then the steady state current in the inductor $I = \frac{6}{6} = 1 \text{ A}$

When T_1 and T_3 are connected then current through inductor remains same. So potential difference across 3Ω
 $V = Ir = 1 \times 3 = 3 \text{ volt}$

Q4

Hints and Solutions

MathonGo

$$|\epsilon| = \frac{d\phi}{dt} = 20t + 20\text{mV}$$

$$|i| = \frac{|\epsilon|}{R} = 10t + 10\text{ mA}$$

at $t = 5$

$$|i| = 60\text{ mA}$$